

Algebraic Geometry 21

Riemann surfaces

Contents

1 Positive Definiteness: Riemannian versus Lorentzian Metrics

One of the most important properties of a metric tensor is *positive definiteness*. This property distinguishes ordinary geometry (Riemannian geometry) from the geometry of space-time (Lorentzian geometry).

The metric tensor determines locally

- lengths,
- angles,
- areas,
- volumes,
- the shape of infinitesimal geometric objects.

1.1 A Small Cube Around a Point

Imagine an infinitesimally small cube centered at a point of a manifold.

```

+-----+
/|      /|
+-----+ |
| |      | |
| +-----|-+
|/      |/
+-----+

```

The metric determines the lengths of the edges and the angles between them.

If the metric varies from point to point, this cube may become

- stretched,
- compressed,
- sheared,
- rotated.

In Riemannian geometry it always remains a genuine geometric object with positive edge lengths and positive volume.

1.2 Metric as an Inner Product

For every tangent vector

$$v = (v_1, v_2, \dots, v_n),$$

the metric computes

$$g(v, v).$$

This quantity plays the role of the squared length of the vector.

1.3 Positive Definiteness

Definition 1.3.1. A symmetric metric tensor is called **positive definite** if

$$g(v, v) > 0$$

for every nonzero vector

$$v \neq 0.$$

Furthermore,

$$g(0, 0) = 0.$$

Thus every nonzero direction has positive squared length.

1.4 Example: Euclidean Space

The standard Euclidean metric is

$$g = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Hence

$$g(v, v) = x^2 + y^2 + z^2.$$

Since

$$x^2 + y^2 + z^2 > 0$$

for every nonzero vector,

the metric is positive definite.

1.5 Stretching the Geometry

Suppose instead

$$g = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 9 \end{pmatrix}.$$

Then

$$g(v, v) = 4x^2 + y^2 + 9z^2.$$

Geometrically,

- the x direction is measured twice as long,
- the z direction three times as long,
- the y direction is unchanged.

The infinitesimal cube becomes an infinitesimal rectangular box.

Original cube

```
+-----+
|       |
|       |
+-----+
```

↓

Metric deformation

```
+-----+
|       |
|       |
|       |
+-----+
```

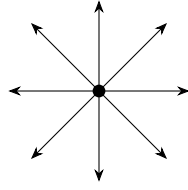
All lengths remain positive.

The local volume remains positive.

Orientation is preserved.

1.6 Why the Name "Positive Definite"?

Imagine trying every possible direction.



For every nonzero arrow,

$$g(v, v) > 0.$$

There are no exceptions.

1.7 Riemannian Metrics

A **Riemannian metric** is a metric tensor satisfying

1. symmetry,

$$g_{ij} = g_{ji},$$

2. positive definiteness.

Consequently,

- every tangent space looks locally like Euclidean space,
- distances are always positive,
- infinitesimal balls become ellipsoids,
- infinitesimal cubes become distorted boxes.

1.8 Example: The Sphere

The unit sphere has metric

$$ds^2 = d\theta^2 + \sin^2 \theta d\phi^2.$$

Thus

$$g = \begin{pmatrix} 1 & 0 \\ 0 & \sin^2 \theta \end{pmatrix}.$$

For every tangent vector

$$v = (v_\theta, v_\phi),$$

we obtain

$$g(v, v) = v_\theta^2 + \sin^2 \theta v_\phi^2.$$

Away from the poles,

$$\sin \theta > 0,$$

therefore

$$g(v, v) > 0$$

for every nonzero tangent vector.

Hence the sphere is a Riemannian manifold.

2 Lorentzian Metrics

Space-time geometry is fundamentally different.

Instead of

$$(+, +, +, +),$$

the metric has signature

$$(-, +, +, +)$$

(or equivalently $(+, -, -, -)$).

The Minkowski metric is

$$g = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

For a vector

$$v = (t, x, y, z),$$

we compute

$$g(v, v) = -t^2 + x^2 + y^2 + z^2.$$

Notice the minus sign.

2.1 Loss of Positive Definiteness

Now three possibilities exist.

Timelike vectors

$$g(v, v) < 0.$$

Example:

$$v = (1, 0, 0, 0),$$

gives

$$g(v, v) = -1.$$

Spacelike vectors

$$g(v, v) > 0.$$

Example:

$$v = (0, 1, 0, 0),$$

gives

$$g(v, v) = 1.$$

Lightlike (null) vectors

$$g(v, v) = 0.$$

Example:

$$v = (1, 1, 0, 0),$$

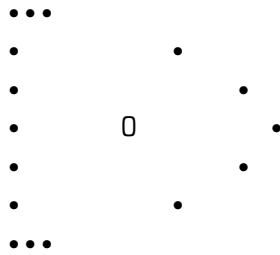
gives

$$g(v, v) = -1 + 1 = 0.$$

Therefore the Lorentz metric is *not* positive definite.

2.2 Geometric Picture

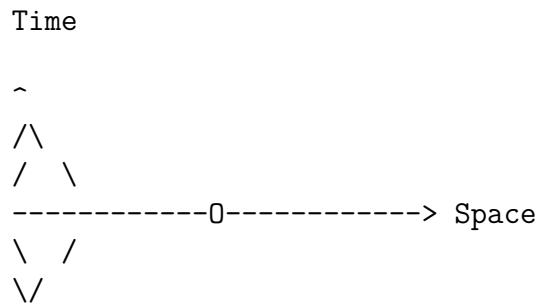
In Euclidean geometry,
vectors of equal length form a sphere.



Every direction behaves in essentially the same way.

In Lorentzian geometry,

the distinguished object is the light cone.



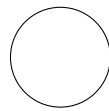
The light cone divides vectors into

- timelike,
- lightlike,
- spacelike.

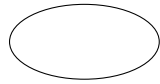
2.3 What Happens to the Infinitesimal Cube?

Riemannian Metric

A small Euclidean ball becomes an ellipsoid.

Euclidean Ball

Metric deformation

**Ellipsoid**

Likewise,

a small cube becomes a skewed or stretched box.

All principal directions have positive lengths.

Lorentzian Metric

There is no ordinary infinitesimal ball.

Instead, the local geometry is organized by the light cone.

Time behaves differently from spatial directions.

Rather than ellipsoids, Lorentzian geometry naturally produces

- light cones,
- hyperboloids,
- causal regions.

3 Comparison

	Riemannian Metric	Lorentzian Metric
Signature	$(+, +, +)$ or $(+, +, +, +)$	$(-, +, +, +)$
Positive definite	Yes	No
Possible values of $g(v, v)$	Always positive for $v \neq 0$	Positive, zero, or negative
Local geometry	Ellipsoids	Light cones and hyperboloids
Distances	Always positive	Space-time intervals
Physical interpretation	Ordinary geometry	General relativity

4 Main Idea

Positive definiteness guarantees that every nonzero direction has a strictly positive squared length,

$$g(v, v) > 0.$$

Therefore all infinitesimal geometric objects remain ordinary Euclidean objects, only stretched or compressed.

Lorentzian geometry intentionally abandons positive definiteness.

The metric distinguishes time from space, allowing

$$g(v, v) < 0, \quad = 0, \quad > 0,$$

which gives rise to the causal structure of spacetime, light cones, timelike curves, and the geometry used in Einstein's theory of general relativity.

5 What Does a Negative Length Mean in a Lorentzian Metric?

One of the most common misconceptions is that a negative value of

$$g(v, v)$$

means a “negative length.”

It does *not*.

Instead, the quantity

$$g(v, v)$$

is called the **Lorentzian norm squared** or the **space–time interval squared**.

Unlike Euclidean geometry, it is not always positive.

5.1 Euclidean Geometry

In Euclidean space,

$$g(v, v) = x^2 + y^2 + z^2.$$

Since this quantity is always positive,

we define the length

$$|v| = \sqrt{g(v, v)}.$$

Everything behaves as expected.

5.2 Lorentzian Geometry

In Minkowski space,

$$g = \text{diag}(-1, 1, 1, 1),$$

so that

$$g(v, v) = -t^2 + x^2 + y^2 + z^2.$$

Notice that the time coordinate contributes negatively.

5.3 Example 1: Standing Still

Suppose

$$v = (1, 0, 0, 0).$$

Then

$$g(v, v) = -1.$$

This does *not* mean that the vector has negative length.

Instead, it means the displacement is entirely through time.

The physically meaningful quantity is the **proper time**

$$d\tau = \sqrt{-g(v, v)}.$$

Thus

$$d\tau = 1.$$

5.4 Example 2: Pure Spatial Motion

Suppose

$$v = (0, 3, 4, 0).$$

Then

$$g(v, v) = 25.$$

The interval is positive and corresponds to an ordinary spatial distance,

$$\sqrt{25} = 5.$$

5.5 Example 3: Light

Suppose

$$v = (5, 5, 0, 0).$$

Then

$$g(v, v) = -25 + 25 = 0.$$

This vector lies on the light cone.

Such vectors are called

- null,
- lightlike,
- isotropic.

5.6 Meaning of the Sign

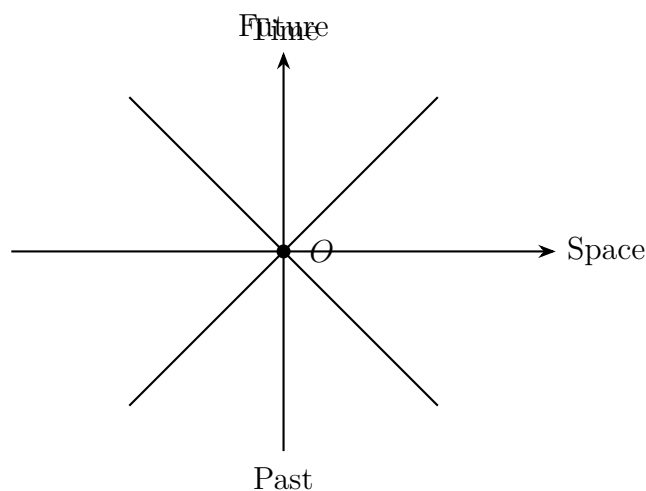
The sign of

$$g(v, v)$$

classifies vectors into three fundamentally different types.

Value of $g(v, v)$	Type	Meaning
$g(v, v) < 0$	Timelike	Motion mainly through time
$g(v, v) = 0$	Lightlike	Motion of light
$g(v, v) > 0$	Spacelike	Separation mainly through space

5.7 Light Cone Interpretation



The light cone divides vectors into

- timelike vectors (inside the cone),
- lightlike vectors (on the cone),
- spacelike vectors (outside the cone).

5.8 Why Is Time Negative?

The negative sign is a matter of convention.

Many books instead use

$$(+, -, -, -).$$

Then

- timelike vectors become positive,
- spacelike vectors become negative.

The geometry and physics remain exactly the same.

5.9 Physical Interpretation

The Lorentz metric compares the contributions of time and space.

Using the convention

$$(-, +, +, +),$$

we compute

$$g(v, v) = -(\text{time})^2 + (\text{space})^2.$$

Therefore

- if the time contribution dominates,

$$g(v, v) < 0,$$

the vector is timelike;

- if the space contribution dominates,

$$g(v, v) > 0,$$

the vector is spacelike;

- if they are exactly equal,

$$g(v, v) = 0,$$

the vector is lightlike.

5.10 Main Idea

In Euclidean geometry the metric answers

“How long is this vector?”

In Lorentzian geometry it answers

“Is this displacement mainly through time, mainly through space, or exactly at the speed of light?”

Thus a negative Lorentzian norm does *not* represent a negative physical length.

Instead it indicates a timelike displacement.

6 Do We Move Through Space and Time at the Same Time?

The answer is

Yes.

Every physical object always moves through both space and time.

The difference is only in the proportion.

6.1 Standing Still

Suppose you are sitting in a chair.

Relative to the Earth,

$$dx = dy = dz = 0,$$

while

$$dt > 0.$$

Your displacement vector is

$$(dt, 0, 0, 0).$$

Therefore

$$ds^2 = -c^2 dt^2.$$

Since

$$ds^2 < 0,$$

your motion is timelike.

Although you are not moving through space, you are continuously moving forward through time.

6.2 Walking

Suppose during one second you walk one metre.

Now

$$dx > 0,$$

and

$$dt > 0.$$

Hence

$$ds^2 = -c^2 dt^2 + dx^2.$$

Because

$$c = 3 \times 10^8 \text{ m/s},$$

the quantity

$$c^2 dt^2$$

is enormously larger than

$$dx^2.$$

Therefore

$$ds^2 < 0.$$

You are still moving mainly through time.

6.3 Travelling Near the Speed of Light

Suppose a spacecraft moves with

$$v = 0.99c.$$

Now

$$dx \approx c dt.$$

Hence

$$-c^2 dt^2 + dx^2$$

is much closer to zero.

The worldline tilts toward the light cone.

The astronaut experiences less proper time.

This is the geometric origin of time dilation.

6.4 Light

For light,

$$dx = c dt.$$

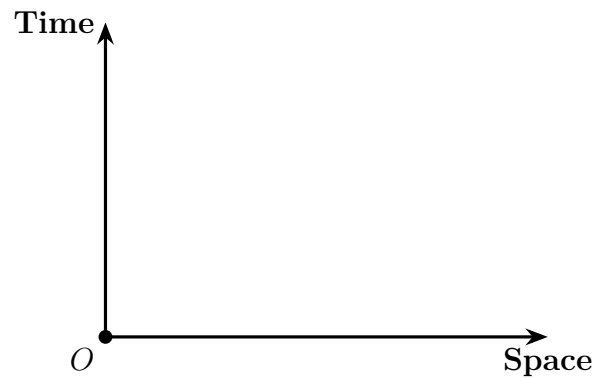
Thus

$$ds^2 = -c^2 dt^2 + c^2 dt^2 = 0.$$

Light always follows null (lightlike) directions.

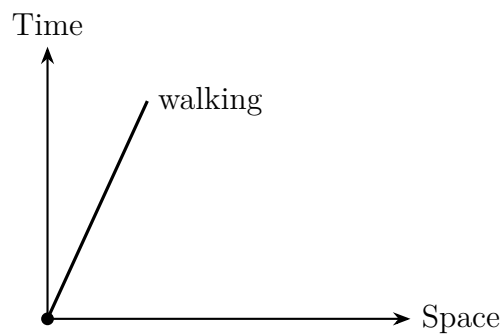
6.5 Geometric Picture

Standing still:



Almost all motion is through time.

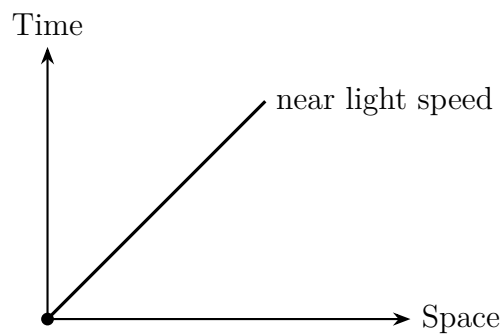
Walking:



Part of the motion is through space.

Part is through time.

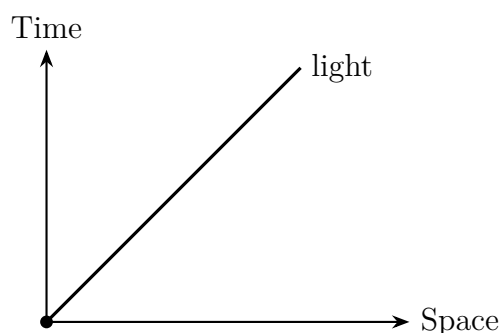
Near light speed:



More motion is through space.

Less proper time elapses.

Light:



Light travels exactly on the light cone.

6.6 Four-Velocity

Every massive particle has a four-velocity

$$U^\mu.$$

Its Lorentzian norm is always

$$g(U, U) = -c^2$$

(using the signature $(-, +, +, +)$).

This remarkable equation means that every massive object moves through spacetime with the same invariant spacetime “speed.”

Different observers disagree about the decomposition into space and time, but they all agree on the invariant norm

$$g(U, U).$$

6.7 Final Interpretation

Every object always moves through spacetime.

The Lorentz metric tells us how that motion is divided between

- motion through time,
- motion through space.

Moving faster through space causes the worldline to tilt toward the light cone.

Consequently,

- less proper time elapses,
- clocks run slower,
- time dilation naturally appears.

This is one of the most beautiful geometric consequences of the Lorentzian metric.

7 How Does the Lorentzian Metric Change in General Relativity?

A very natural question is:

If mass curves space–time, does this simply mean that the off-diagonal entries of the Lorentzian metric become nonzero?

The answer is

Sometimes yes, but not always.

There are two independent ways in which the metric may differ from the flat Minkowski metric.

7.1 Flat Space–Time (Special Relativity)

In special relativity the metric is the same everywhere.

The Minkowski metric is

$$g_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Notice that

- all diagonal entries are constant,
- every off-diagonal entry is zero,
- every point of space–time has exactly the same metric.

Graphically,

Every point of spacetime carries the same

$$(-, +, +, +)$$

Lorentzian signature

There is

- no curvature,
- no gravity,
- no preferred location.

8 General Relativity

Einstein's revolutionary idea was

Mass and energy do not change the Lorentzian signature.

Instead, they change the metric coefficients.

Instead of a constant matrix,

$$g_{\mu\nu},$$

we now have

$$g_{\mu\nu}(x),$$

whose entries depend on the point

$$x = (t, x, y, z).$$

8.1 Case 1: The Diagonal Entries Change

The most common situation is that only the diagonal coefficients change.

A famous example is the Schwarzschild metric,

$$g_{\mu\nu} = \begin{pmatrix} -\left(1 - \frac{2GM}{rc^2}\right) & 0 & 0 & 0 \\ 0 & \left(1 - \frac{2GM}{rc^2}\right)^{-1} & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin^2 \theta \end{pmatrix}.$$

Notice

All off-diagonal entries are still zero.

Nevertheless,

space-time is already curved.

8.2 What Changes?

Compared with flat space,

the diagonal entries are no longer

$$(-1, 1, 1, 1).$$

Instead they depend on

$$r.$$

Therefore

- clocks run at different speeds,
- measured distances change,
- light bends,
- free particles follow curved geodesics.

Thus curvature does not require nonzero off-diagonal terms.

8.3 Case 2: Off-Diagonal Entries Become Nonzero

A second possibility is that some off-diagonal entries appear.

For example,

around a rotating body,

$$g_{t\phi} \neq 0.$$

Schematically,

$$g_{\mu\nu} = \begin{pmatrix} -0.8 & 0 & 0 & -a \\ 0 & 1.2 & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ -a & 0 & 0 & \dots \end{pmatrix}.$$

where

$$g_{t\phi} = g_{\phi t} = -a.$$

8.4 Meaning of Off-Diagonal Entries

Recall that

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu.$$

If the metric is diagonal,

$$ds^2 = g_{tt} dt^2 + g_{xx} dx^2 + g_{yy} dy^2 + g_{zz} dz^2.$$

Every coordinate contributes independently.

Suppose instead

$$g_{tx} \neq 0.$$

Then

$$ds^2 = g_{tt} dt^2 + 2g_{tx} dt dx + g_{xx} dx^2 + \dots$$

Now

time and the spatial coordinate

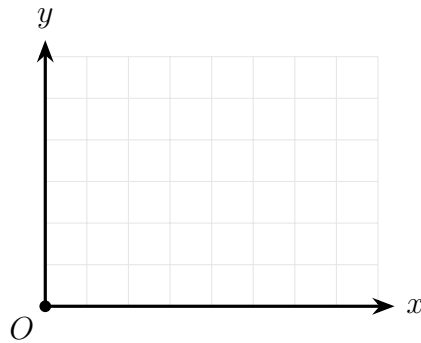
$$x$$

become coupled.

Moving in one direction affects measurements in the other.

8.5 Euclidean Analogy

Suppose we have ordinary Cartesian coordinates.

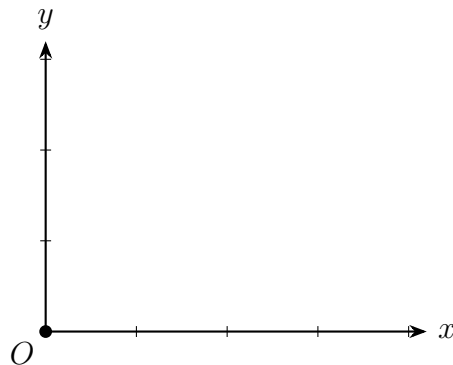


The metric is

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

The coordinate axes are perpendicular.

Now suppose we choose skew coordinates.



Now the metric becomes

$$\begin{pmatrix} 1 & a \\ a & 1 \end{pmatrix},$$

where

$$a \neq 0.$$

The off-diagonal entries indicate that the basis vectors are no longer orthogonal.

8.6 Lorentzian Interpretation

Exactly the same phenomenon occurs in General Relativity.

The only difference is that one coordinate is time.

If

$$g_{t\phi} \neq 0,$$

then

time and rotation become geometrically coupled.

This produces

- frame dragging,
- rotational effects,
- twisting of inertial frames.

9 Mass Does Not Change the Signature

One of Einstein's deepest discoveries is

Mass changes the metric, but not its Lorentzian signature.

At every point of space–time,

the metric still has

$$(-, +, +, +),$$

meaning

- one timelike direction,
- three spacelike directions.

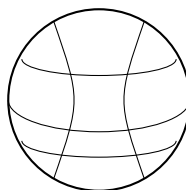
Thus every infinitesimal neighbourhood still looks locally like Minkowski space.

10 Local Flatness

An excellent analogy is the Earth's surface.

Globally,

the Earth is curved.



Earth

However,
 standing in your garden,
 everything appears flat.
 Likewise,
 space–time may be globally curved,
 yet every sufficiently small neighbourhood admits coordinates in which

$$g_{\mu\nu} \approx \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

This is the principle of *local flatness*.

11 Einstein's Equation

Matter tells space–time how to curve.

Space–time tells matter how to move.

Einstein's field equation is

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}.$$

Here

- $T_{\mu\nu}$ is the energy–momentum tensor,
- $G_{\mu\nu}$ is the Einstein curvature tensor,
- the unknown quantity is the metric

$$g_{\mu\nu}(x).$$

12 Comparison

	Special Relativity	General Relativity
Metric coefficients	Constant	Depend on position
Diagonal entries	Constant	Usually vary
Off-diagonal entries	Zero	May be zero or nonzero
Curvature	Zero	Usually nonzero
Gravity	Absent	Produced by curved metric
Signature	$(-, +, +, +)$	Still $(-, +, +, +)$ locally
Local geometry	Exactly Minkowski	Infinitesimally Minkowski

13 Main Ideas

The key facts are

1. The Lorentzian signature never changes.
2. Mass and energy modify the metric coefficients $g_{\mu\nu}(x)$.
3. Curvature already appears when only the diagonal entries vary.
4. Nonzero off-diagonal entries represent couplings between different coordinate directions.
5. In rotating space-times, these couplings produce phenomena such as frame dragging.
6. Every sufficiently small neighbourhood of space-time is locally indistinguishable from flat Minkowski space.

The Lorentzian metric therefore remains the fundamental local geometry of space-time, while gravity is encoded in the way its coefficients vary from point to point.

“`latex`

14 Lorentzian Signature vs. Lorentzian Metric

One of the most important distinctions in differential geometry and general relativity is between a *Lorentzian signature* and a *Lorentzian metric*. These notions are closely related but are not identical.

14.1 The Signed Identity Matrix

Consider the matrix

$$\eta = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

This matrix is called the **Minkowski metric** (written in standard inertial coordinates). It is

- a Lorentzian metric,
- has Lorentzian signature,
- is constant throughout spacetime,
- represents flat spacetime.

It is the simplest example of a Lorentzian metric.

Notice that this matrix is *one particular metric*; it is not the definition of every Lorentzian metric.

14.2 Lorentzian Signature

A metric tensor is called **Lorentzian** if, at every point of the manifold, its matrix has

one negative eigenvalue and three positive eigenvalues

(or equivalently one positive and three negative, depending on convention).

Thus the signature is

$$(-, +, +, +),$$

or equivalently

$$(+, -, -, -).$$

The signature specifies only the signs of the eigenvalues.

It does *not* determine the numerical values of the entries of the metric tensor.

14.3 Many Different Lorentzian Metrics

There are infinitely many matrices having Lorentzian signature.

For example,

$$\begin{pmatrix} -2 & 0 & 0 & 0 \\ 0 & 5 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 7 \end{pmatrix}$$

is Lorentzian because its eigenvalues are

$$(-2, 5, 1, 7),$$

which contain exactly one negative value.

Likewise,

$$\begin{pmatrix} -1 & 0.3 & 0 & 0 \\ 0.3 & 2 & 0 & 0 \\ 0 & 0 & 4 & 1 \\ 0 & 0 & 1 & 3 \end{pmatrix}$$

is also Lorentzian provided its eigenvalues still consist of

$$1 \text{ negative} + 3 \text{ positive.}$$

Notice that off-diagonal entries are perfectly allowed.

14.4 Examples of Lorentzian Metrics

1. Minkowski Metric

Flat spacetime:

$$g = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Properties:

- flat,
- constant,
- zero curvature,
- describes special relativity.

2. Schwarzschild Metric

Outside a non-rotating spherical body,

$$g = \begin{pmatrix} -\left(1 - \frac{2M}{r}\right) & 0 & 0 & 0 \\ 0 & \left(1 - \frac{2M}{r}\right)^{-1} & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin^2 \theta \end{pmatrix}.$$

Unlike the Minkowski metric,

- the entries depend on position,
- spacetime is curved,
- gravity appears through the changing coefficients.

3. Kerr Metric

A rotating black hole has metric

$$g = (g_{\mu\nu}(x)).$$

Unlike Schwarzschild,

$$g_{t\phi} \neq 0,$$

which represents the coupling between time and angular direction caused by frame dragging.

4. FLRW Metric

In cosmology,

$$g = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & a(t)^2 & 0 & 0 \\ 0 & 0 & a(t)^2 & 0 \\ 0 & 0 & 0 & a(t)^2 \end{pmatrix}.$$

Here

$$a(t)$$

is the scale factor describing the expansion of the Universe.

14.5 General Lorentzian Metric

In differential geometry, the metric is not a single matrix.

Instead,

$$g = (g_{ij}(p)),$$

where

$$p \in M$$

is a point of the manifold.

Thus every point has its own matrix

$$g(p).$$

The entries may change continuously from point to point.

The only requirement is

every matrix $g(p)$ has Lorentzian signature.

14.6 Analogy with Riemannian Metrics

A Riemannian metric satisfies

$$(+, +, +, +),$$

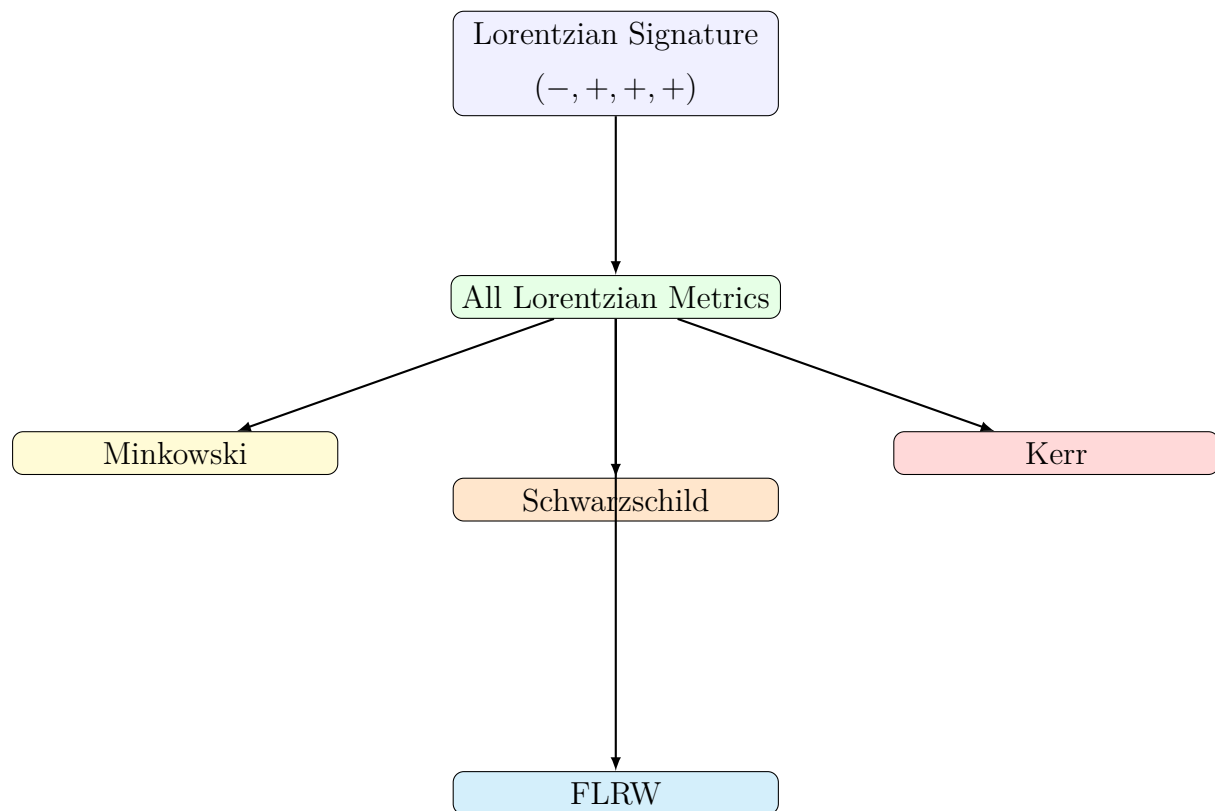
meaning all eigenvalues are positive.

Again, there are infinitely many examples:

- Euclidean metric,
- sphere metric,
- hyperbolic metric,
- metrics on arbitrary manifolds.

Each has different entries while sharing the same positive-definite signature.

14.7 Visualization



The diagram emphasizes that one signature defines an entire family of Lorentzian metrics.

14.8 Comparison Table

Property	Riemannian	Lorentzian
Signature	$(+, +, +, +)$	$(-, +, +, +)$
Positive definite	Yes	No
Flat example	I	$\eta = \text{diag}(-1, 1, 1, 1)$
Curved examples	Sphere, Hyperbolic	Schwarzschild, Kerr, FLRW
Entries may vary	Yes	Yes
Off-diagonal terms allowed	Yes	Yes
Infinitely many metrics	Yes	Yes

14.9 Key Takeaway

The matrix

$$\eta = \text{diag}(-1, 1, 1, 1)$$

is **not merely a Lorentzian signature**.

It is a specific Lorentzian metric—the **Minkowski metric**.

The term *Lorentzian metric* refers to the entire class of metrics whose matrices have Lorentzian signature at every point.

Therefore,

One Lorentzian signature $\implies$ Infinitely many Lorentzian metrics.

Different Lorentzian metrics describe different spacetimes and different gravitational fields while preserving the same fundamental distinction between one timelike direction and three spacelike directions.